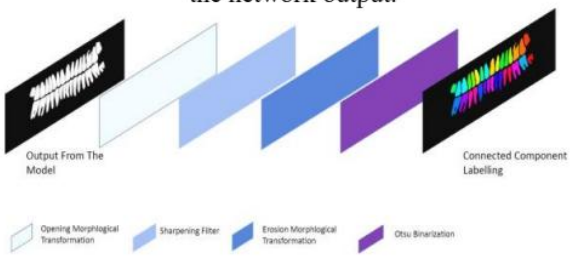
to remove small details and separate teeth using a 5x5 square-shaped structural element. Subsequently, an image sharpening filter is employed to enhance details, followed by grayscale erosion morphological operation applied twice to detach connected teeth. Masks are then generated using Otsu's method for segmentation, and connected components are uniquely labeled with a cluster size threshold of 2000 pixels, ensuring accurate tooth separation and reducing noise in the final segmentation maps.

Fig 3: Steps of the post-processing that are applied to the network output.



IV.EXPERIMENT AND RESULTS

Five experiments are performed to determine the performance of the proposed method under various parameters including different output labels, data augmentation or post processing scenario. In these experiments, a 10-fold cross validation technique is applied. The comparison of the performed procedures is given in Table 1. Whenever the post-processing steps are not applied, segmentation masks are obtained by directly binarizing the sigmoid output of the network using a threshold level of 0.2.

	Sensitivity	*Specificity*	*PPV*	*NPV*
E1	96.6±0.9	98.4±0.4	92.6±1.9	99.3±0.2
E2	96.3±0.6	98.7±0.3	93.9±1.0	99.2±0.1
E3	96.1±0.6	98.8±0.3	94.4±0.8	99.2±0.1
E4	95.9±0.6	98.9±0.2	94.8±0.8	99.1±0.2
E5	99.1±0.3	98.6±0.2	91.9±1.2	99.9±0.0

	Jaccard	*Dice*
E1	89.7±1.2	94.5±0.7
E2	90.6±0.8	95.1±0.5
E3	90.9±0.6	95.2±0.3
E4	91.1±0.5	95.4±0.3
E5	91.1±1.2	95.3±0.6

Table 1: Evaluation metrics

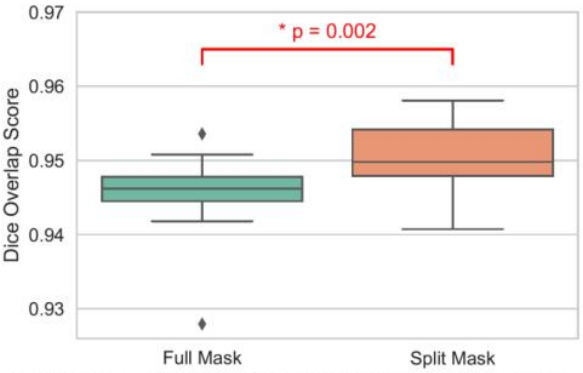


Fig 4: Comparison of the segmentation performance of the network on the full mask (E1) and on the split mask (E2), without any augmentation nor post processing

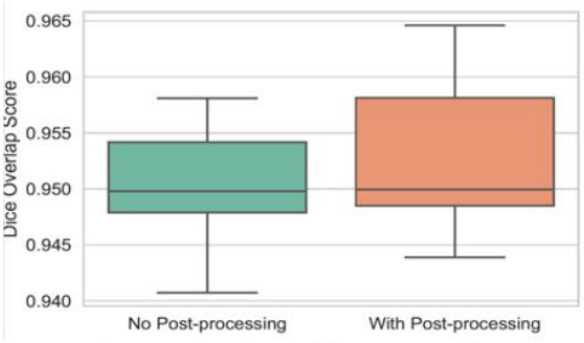


Fig 5: Comparison of the segmentation performance after post-processing with the proposed stages (E5) and without post-processing (E2).

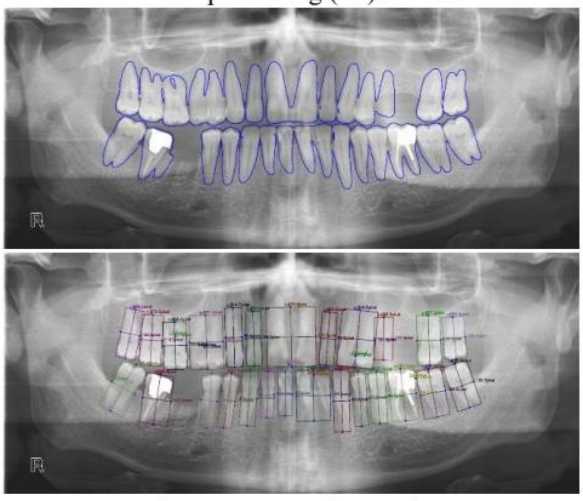


Fig 6: Segmentation results